



ROOM AND PILLAR METHOD OF MINING

FIRST WORKINGS

- First workings are conducted for two main reasons:
 - To develop the pillars, bords and/or panels to enable their subsequent extraction, and
 - To develop multiple main entries for access to the coal reserves. These entries are supported in the long term by the pillars consequently formed.

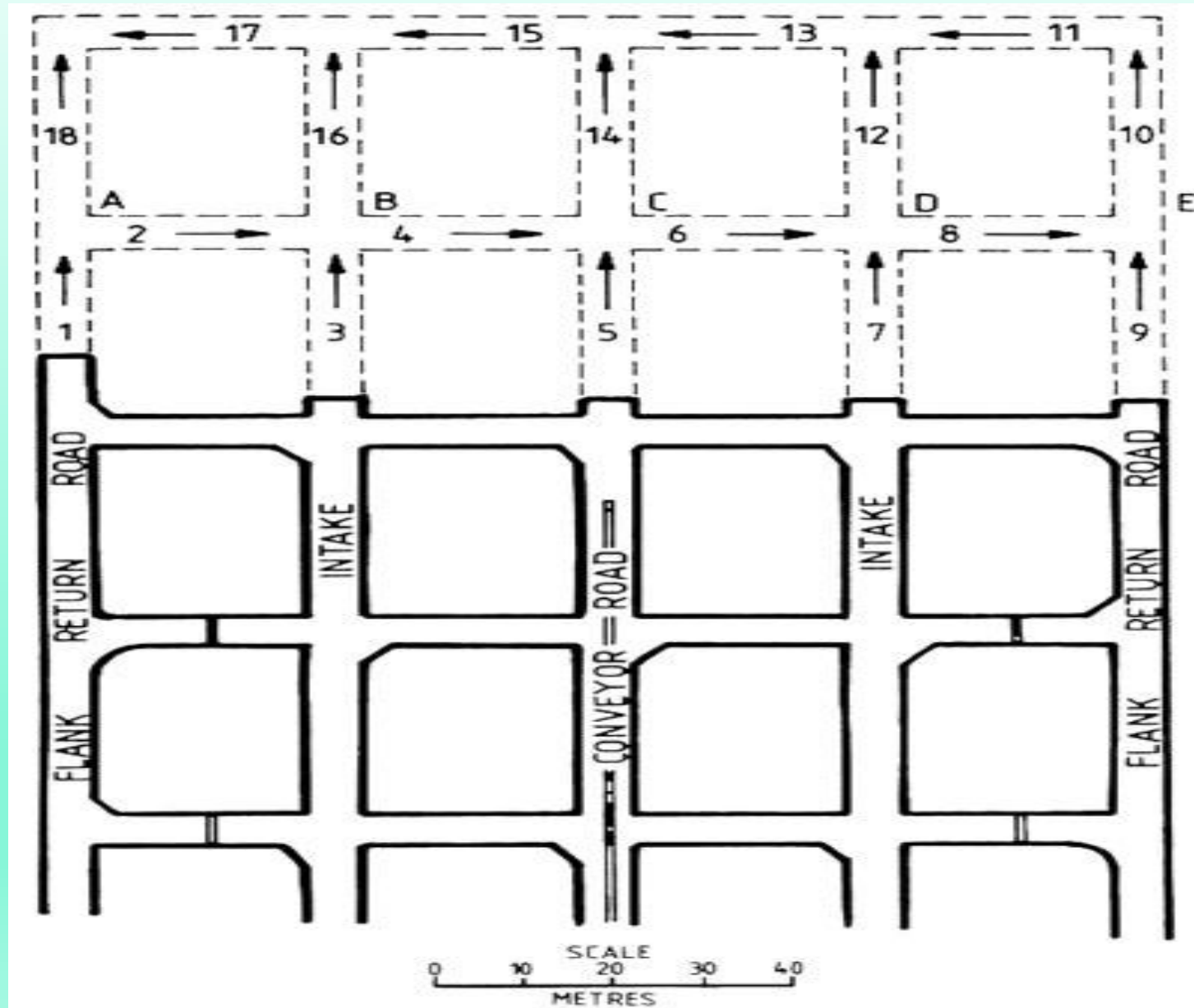
FIRST WORKINGS

The number of development headings driven in any area of workings can vary between two and eight.

This is dependent primarily on the purpose of development, and the second method of working together with the following factors:

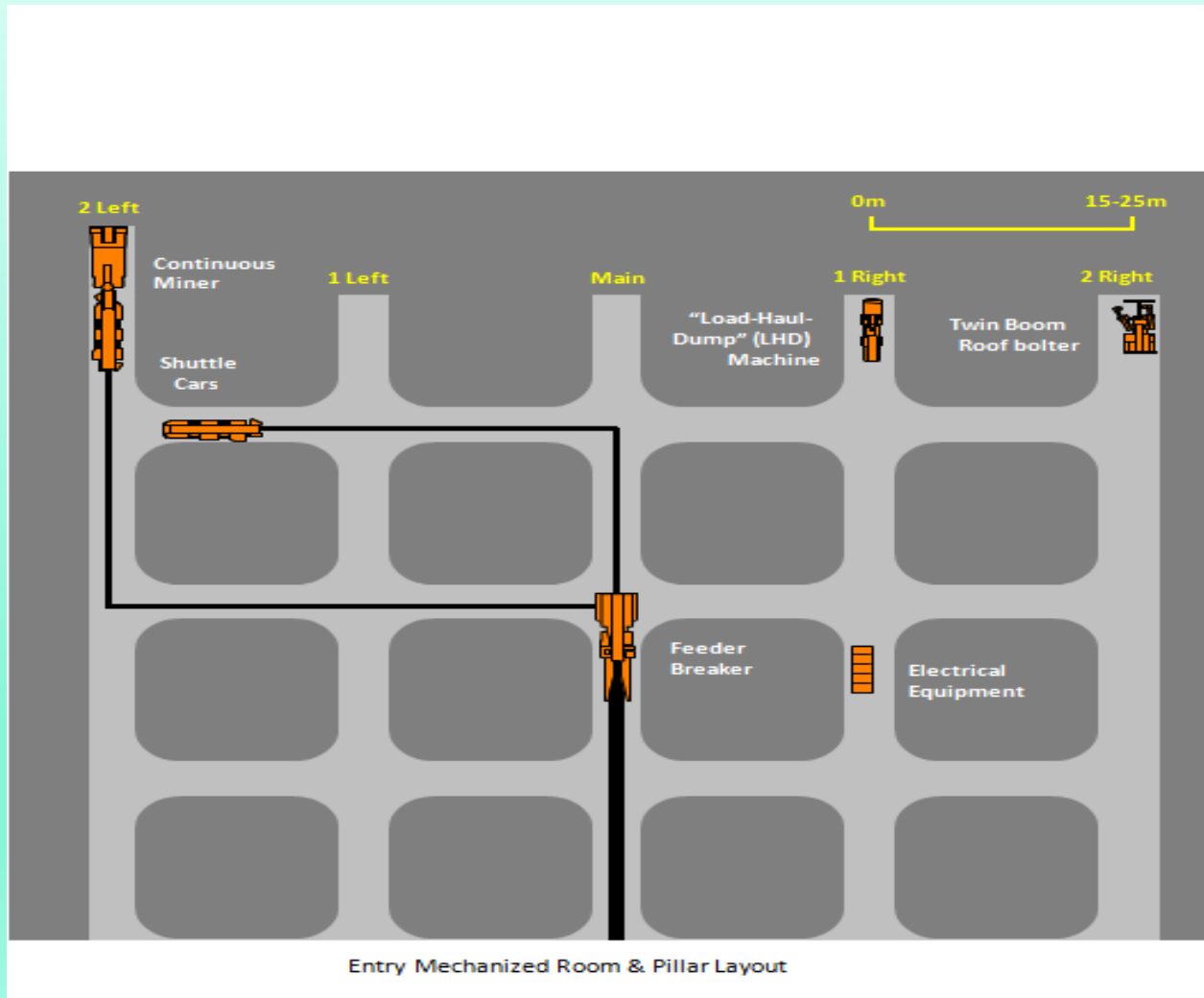
- Mine capacity,
- Number of continuous miner units operating,
- Coal clearance system,
- Men and material transport, and
- Ventilation.

FIRST WORKINGS



Typical Development Layout

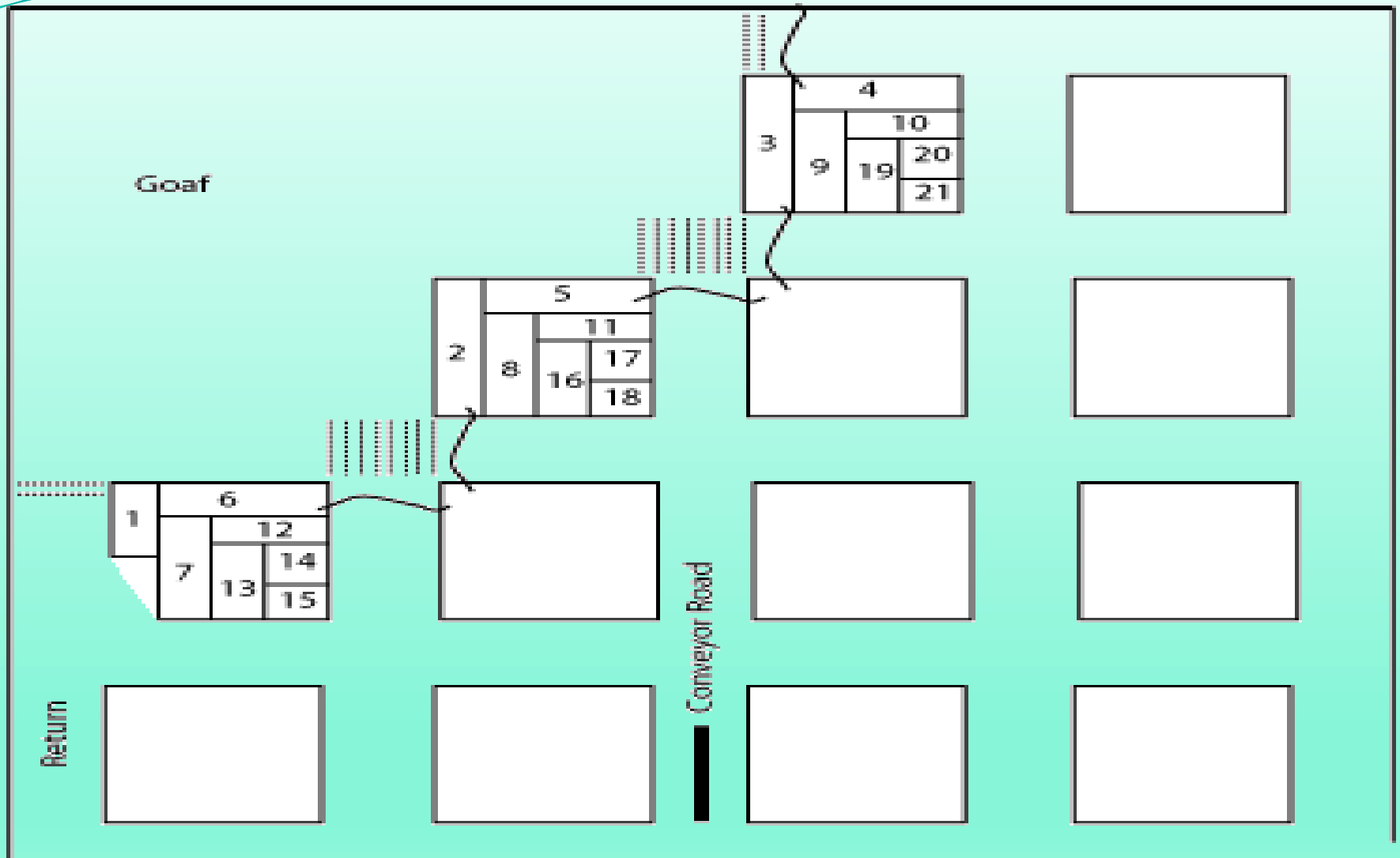
The standard layout consists of a five road development, with the main conveyor and feeder breaker positioned in the central roadway. This provides working areas to carry out all the main development functions:



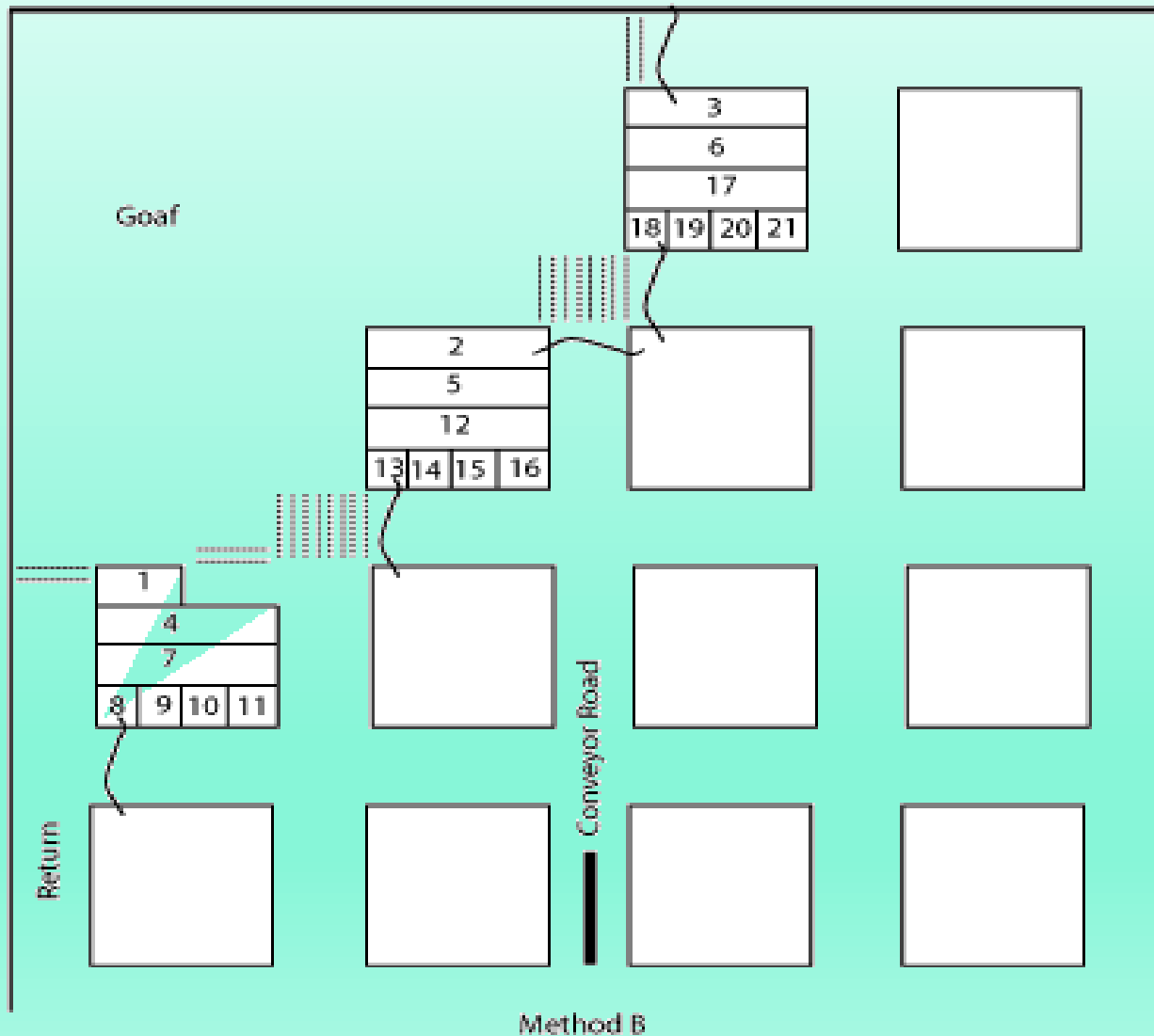
- One heading for the CM to be cutting
- One heading ready for cutting
- One heading for ventilation rearrangement and survey work
- One heading for cleaning by LHD
- One heading for supporting by Quadbolter



OPEN ENDED LIFTING – METHOD A

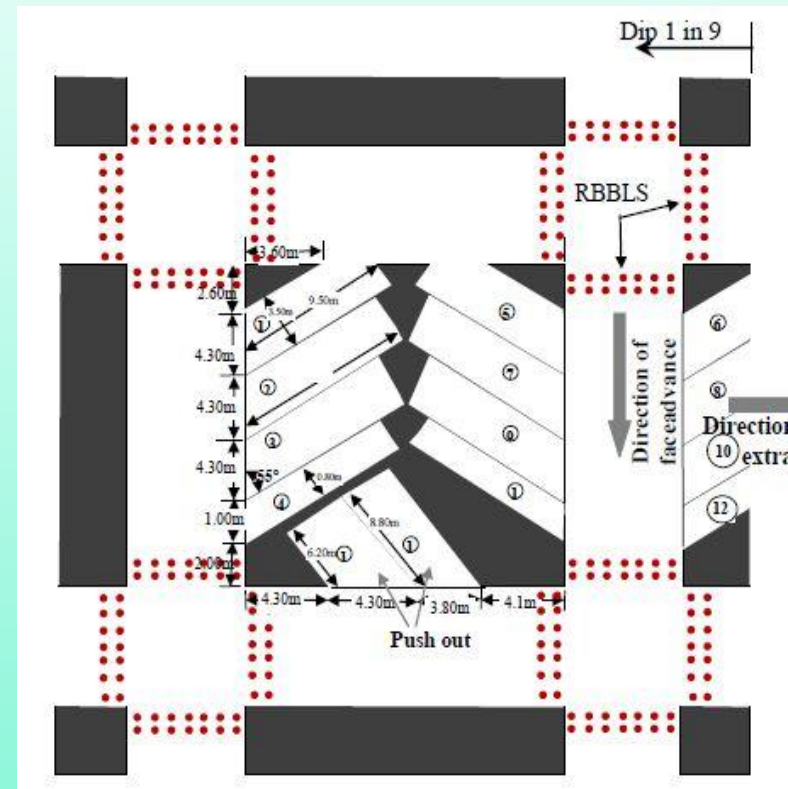


OPEN ENDED LIFTING – METHOD B

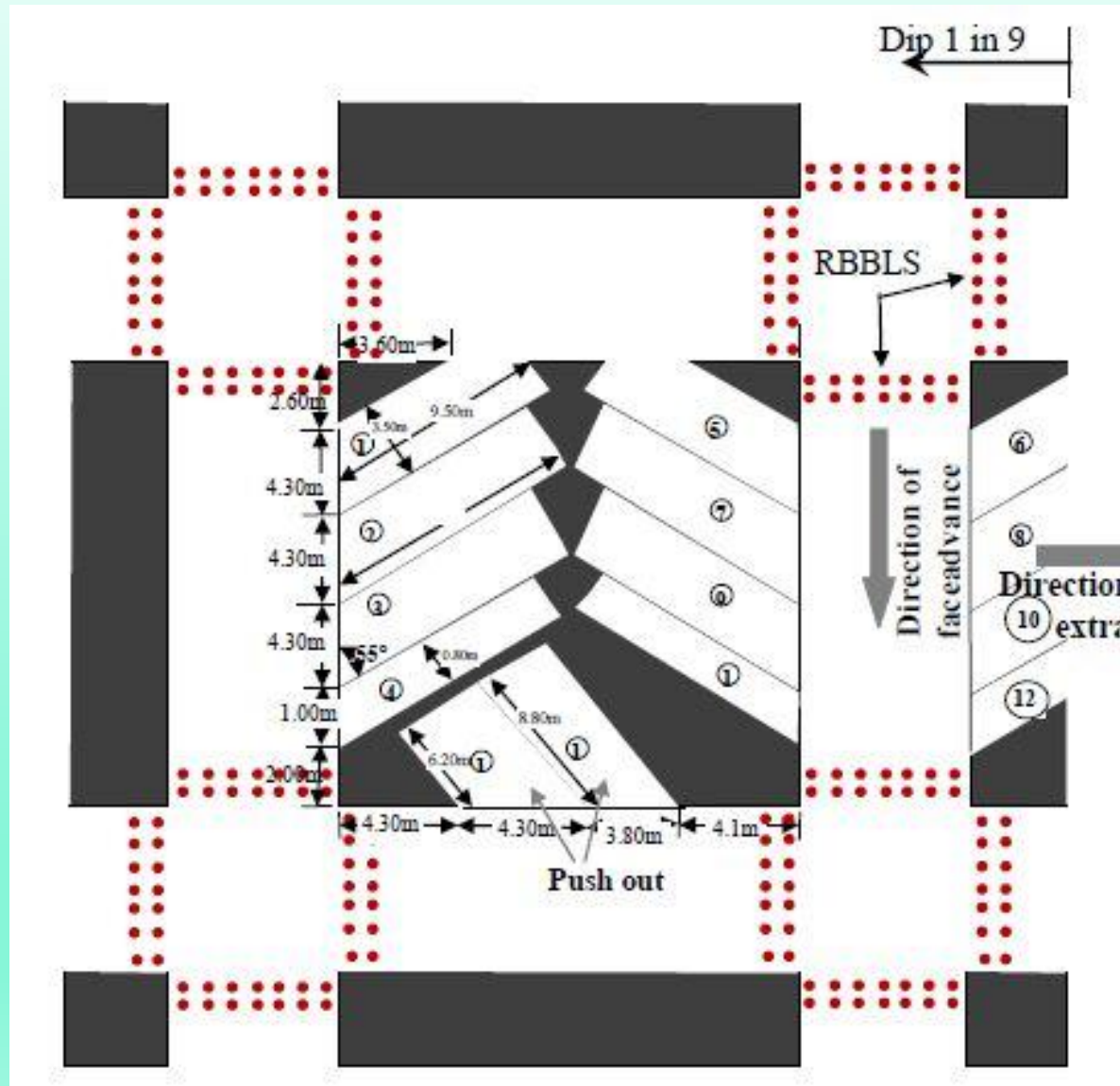


Christmas tree

- Christmas tree method is also known as Fish and Tail or Left-Right method
- Coal is extracted with the help of Continuous Miner by cutting slices into the pillar from both level galleries and then extracting pushouts in the dip direction.
- The first slice is made with a width 3.50m and length 9.50m by leaving a portion of coal 2.60 m towards edge.
- Similarly, two more slices are extracted in the same manner.
- The fourth slice is extracted with a width of 0.80m.
- A straight line of extraction is being maintained and the sequence so followed is to facilitate the caving of roof in dip direction.
- Similar manner of extraction is followed to extract slices on opposite side of previous slices, now push outs are being driven with lengths 6.20m and 8.80m.



Christmas tree method



SPLIT AND FENDER

This is the most common method of pillar mining.

The pillars are usually square or rectangular with dimensions dependent on the depth of working. Typically a pillar dimension of 45 x 45 meter is used at depth of around 450 m.

Basically the pillars developed by the first working are split into sub-pillars and the fenders are then lifted (mined).

The width of the lift is governed by the length of the continuous miner, roof conditions and support rules of the individual mine.

Generally the split is supported by roof bolts or other means of supporting, depending on the timing of the mining.

Roof bolting and W-straps were supplemented with timber props and bars.

SPLIT AND FENDER

The lifting operation took place usually towards the goaf side of the pillar, with the basic philosophy of working adjacent to a competent rib.

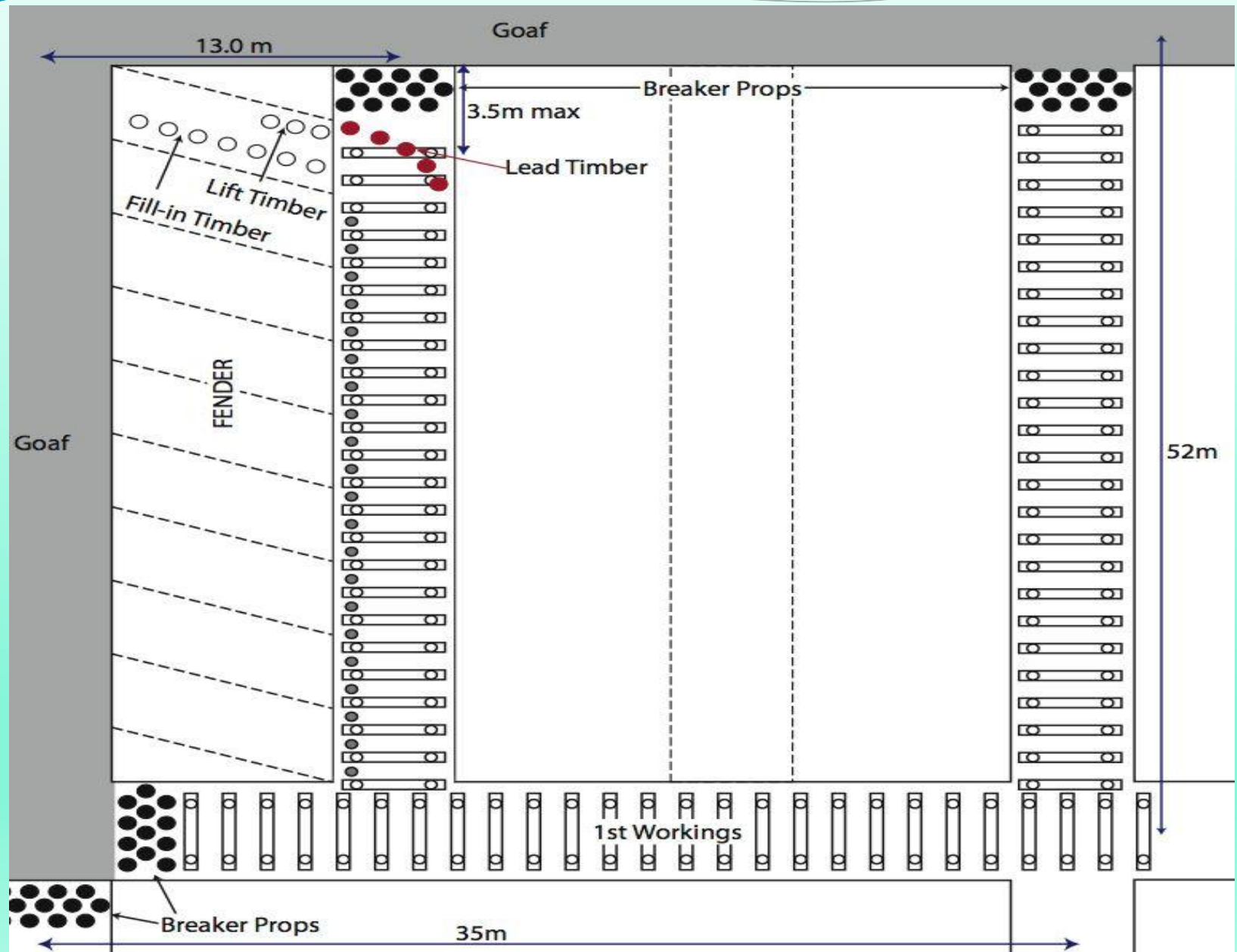
However, in mines with competent roof conditions, lifting from both sides of the split was possible, and with the approval of mine inspectorate.

When extracting on both left and right, pillars up to four meters in width were left in the fenders to provide temporary support.

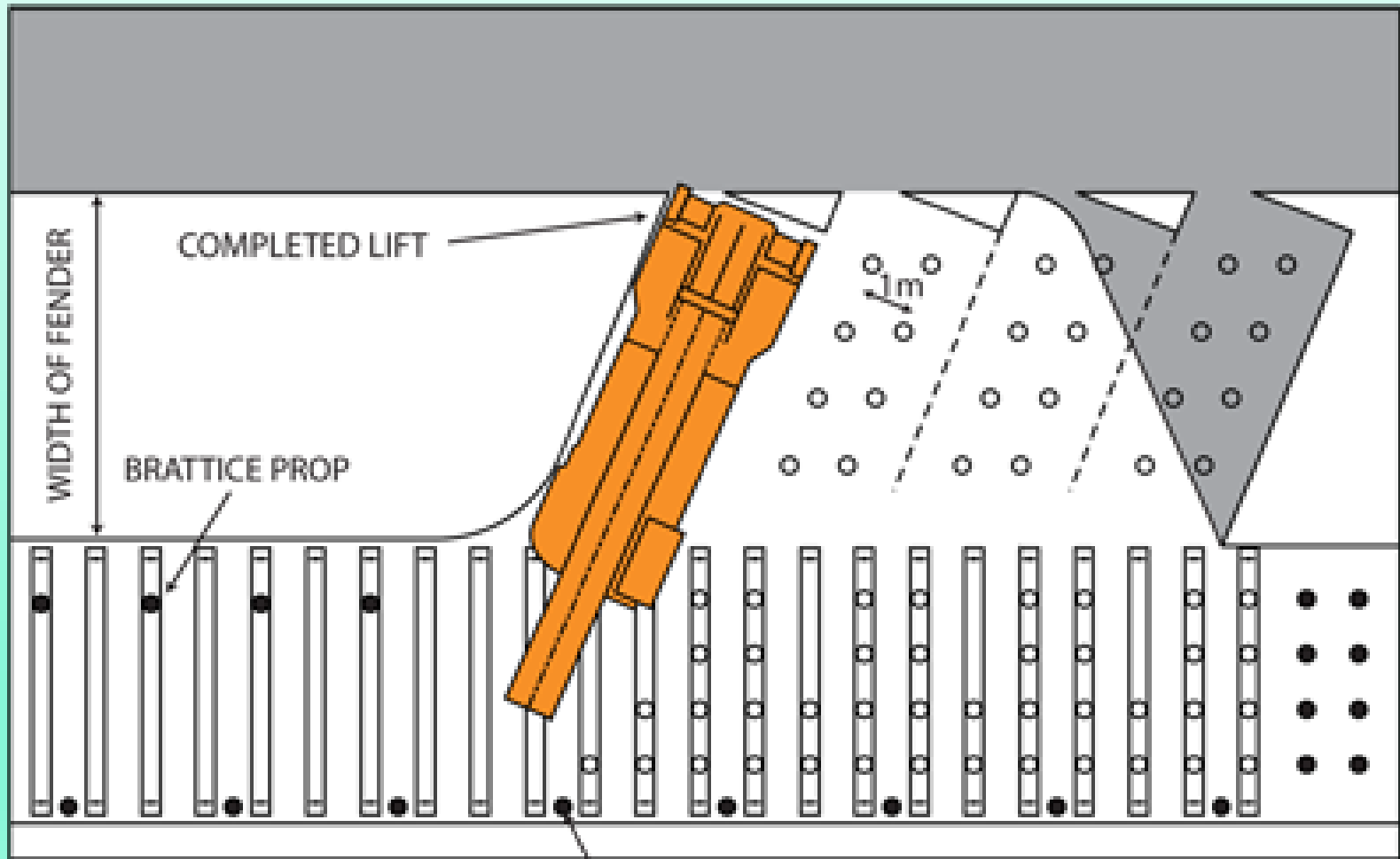
However, the current system of splitting with mobile breaker line supports, lifting from both sides of the split is standard practice.

A small stook (support Buttress) of coal seam (min width 2.0m) is normally left to control caving, particularly on the goaf side.

SPLIT AND FENDER

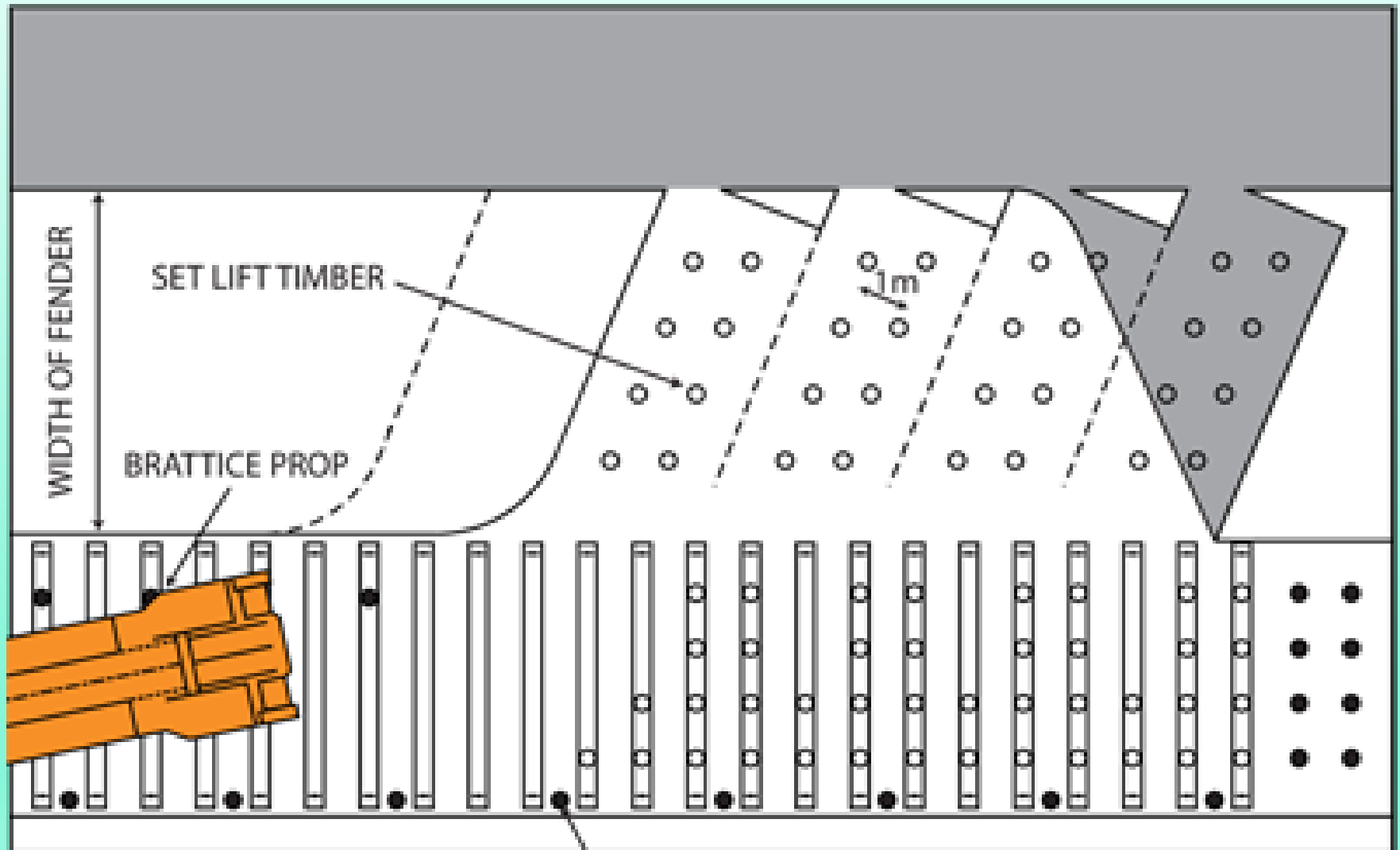


SPLIT AND FENDER



NOTE: SPLITS AT 12.5m CENTRES

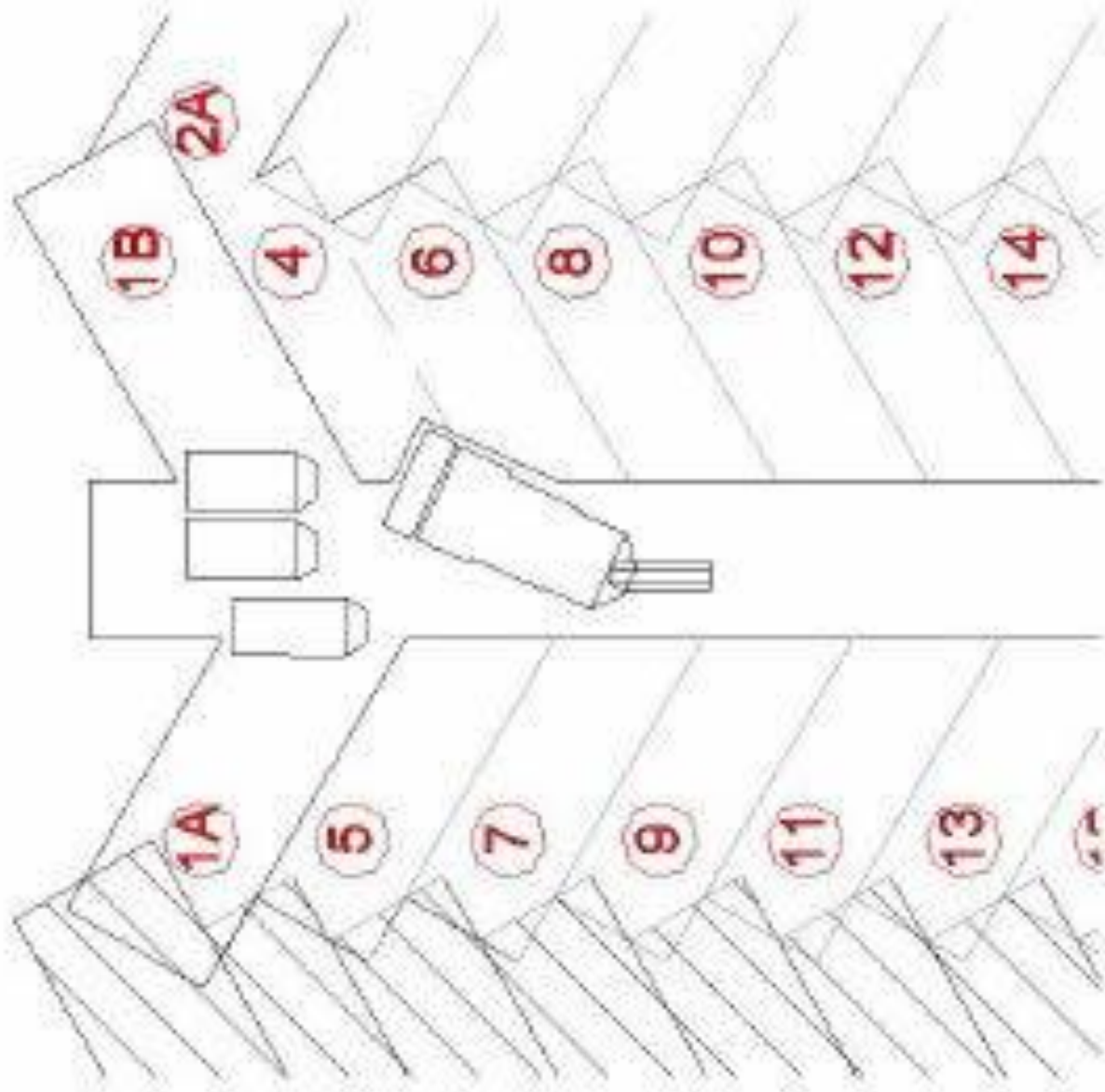
PROP OR 0.9m BOLT @ 3m MAX



NOTE: SPLITS AT 12.5m CENTRES

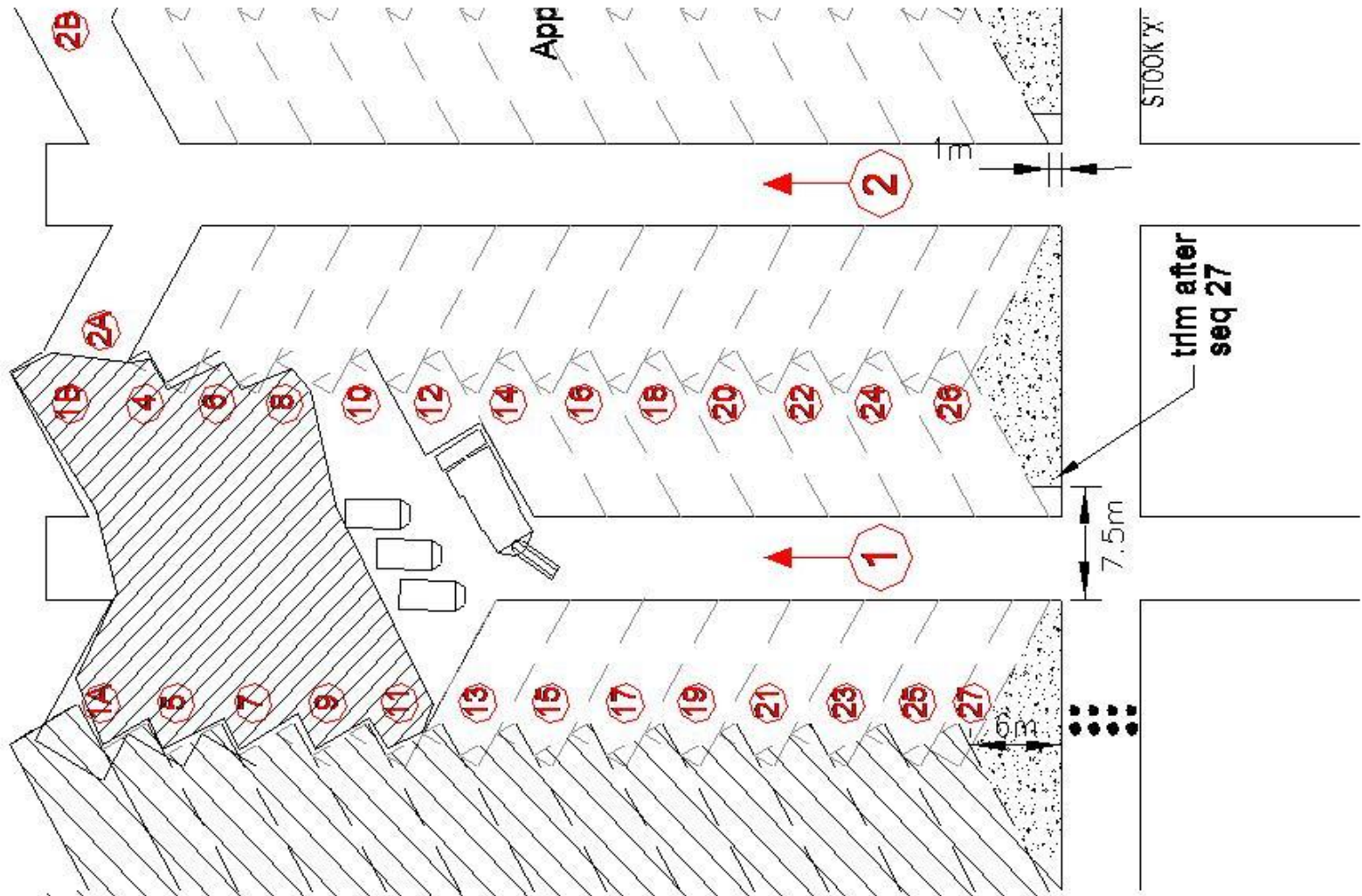
PROP OR 0.9m BOLT @ 3m MAX

SPLIT AND LIFT USING BREAKER LINE SUPPORTS

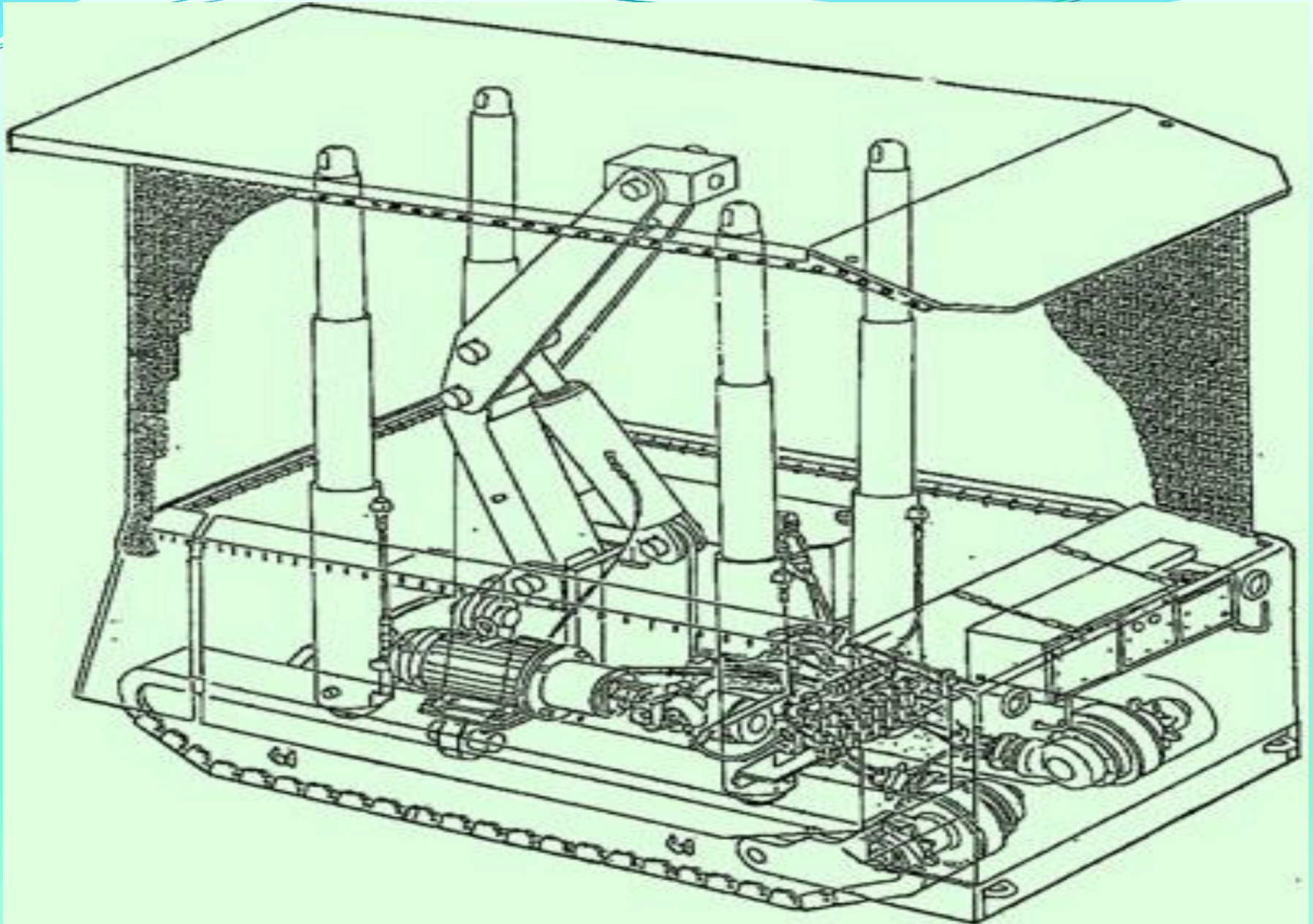


SPLIT AND LIFT USING BREAKER LINE SUPPORTS

GOAF FALLS



MOBILE BREAKER LINE SUPPORT



BREAKER LINE SUPPORT (BLS)

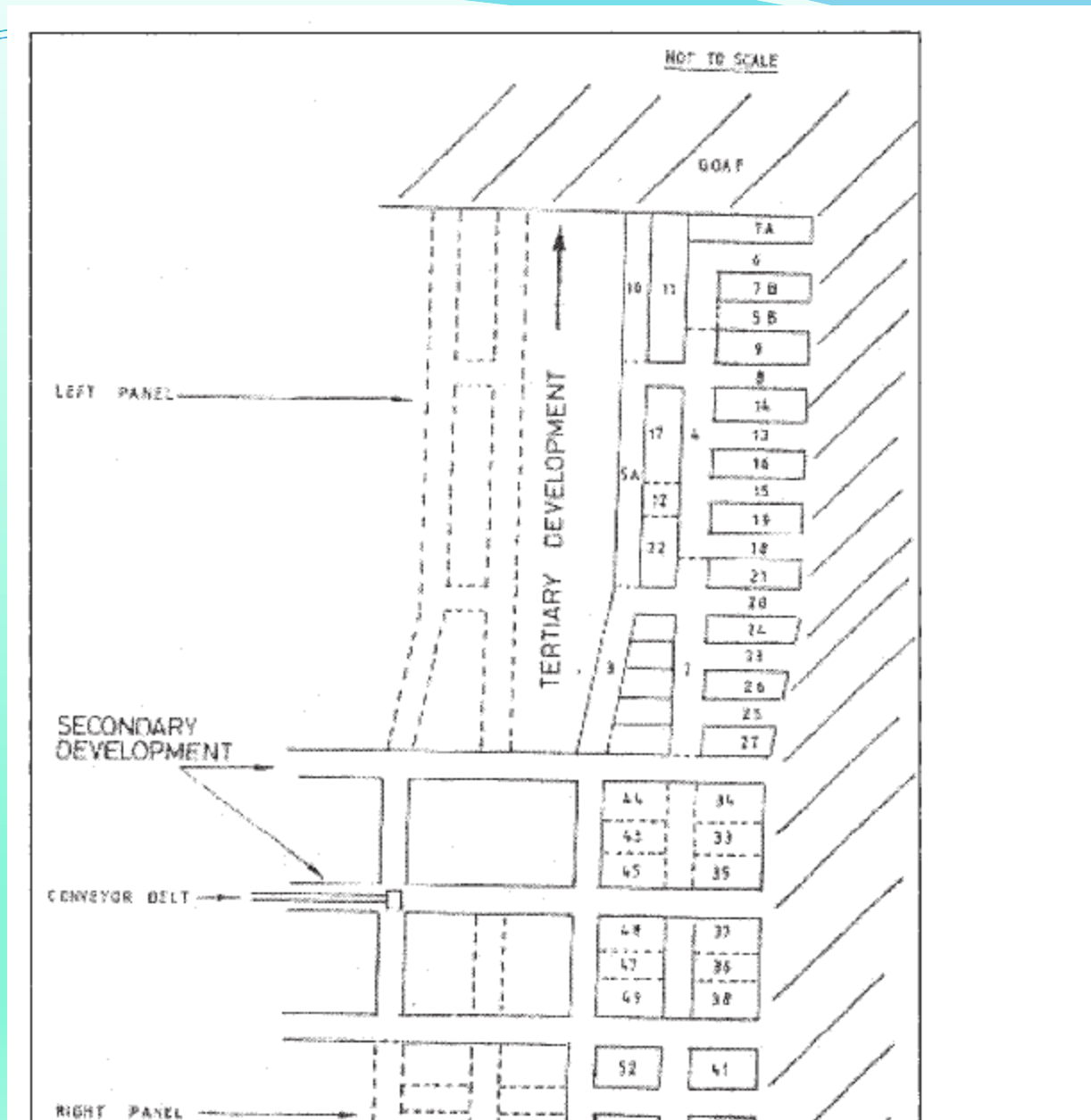
Advantages.

- Greater positive setting loads and roof support on goaf edge .
- Cost saving in timber usage to defray their capital cost.
- Able to be reset to the new breaker line position as extraction takes place faster than timber breaker lines can be set.
- Increases potential continuous miner cutting time and productivity.
- Through remote control they remove the active goaf edge area personnel who would be required otherwise to set the breaker-line props thus further improving safety.

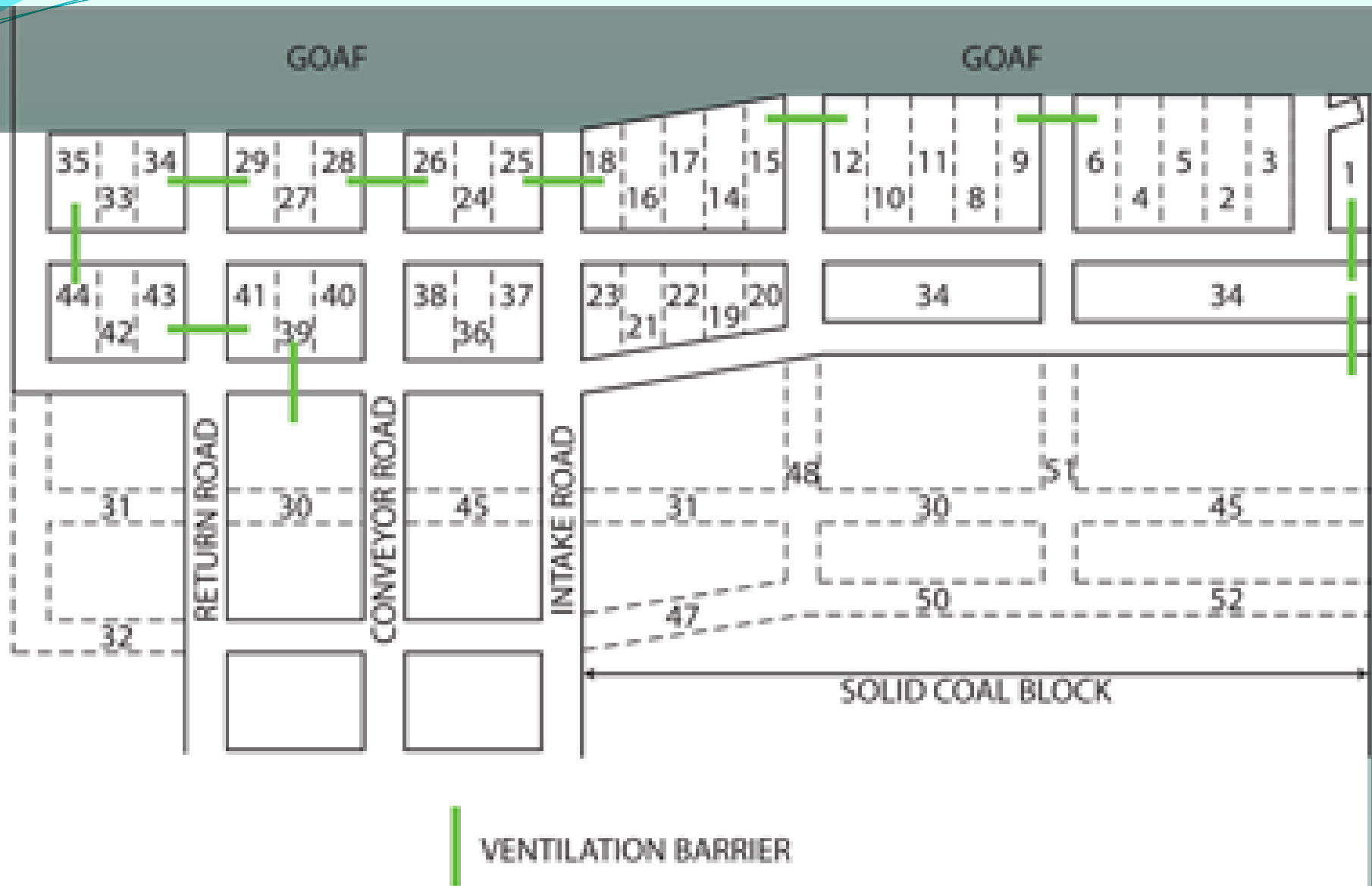
OLD BEN METHOD – PANEL AND PILLAR EXTRACTION

- Developing multiple headings, usually three, to the limits of the area to be extracted.
- These panels, along with newly formed pillars (in the virgin coal) were extracted in retreat.
- Here, the secondary development consisted of three roads, leaving reserves for pillar extraction on either side.
- The total panel width was greater than 200 m.
- Tertiary development, consisting of three roads, was done towards the end of the panel to increase overall extraction.
- From this development, short fenders were then developed and extracted.

OLD BEN METHOD – PANEL AND PILLAR EXTRACTION



OLD BEN METHOD – PANEL AND PILLAR EXTRACTION



WONGAWILLI METHOD

- The Wongawilli system was developed at Wongawilli Colliery in the southern coalfield of New South Wales in the early 1960s.
- Generally with the Wongawilli system, a panel is created by a secondary development consisting of three to five roads and leaving a continuous pillar of coal between the development and the previously caved area.
- The pillar is normally between 50 m and 150 m wide and is extracted by developing and extracting 7 m wide ribs in a modified split and lift system.
- The pillars formed by the development are extracted as the rib extraction retreats. As a result of the length of the rib pillars, this method resembles a shortwall face.

WONGAWILLI METHOD

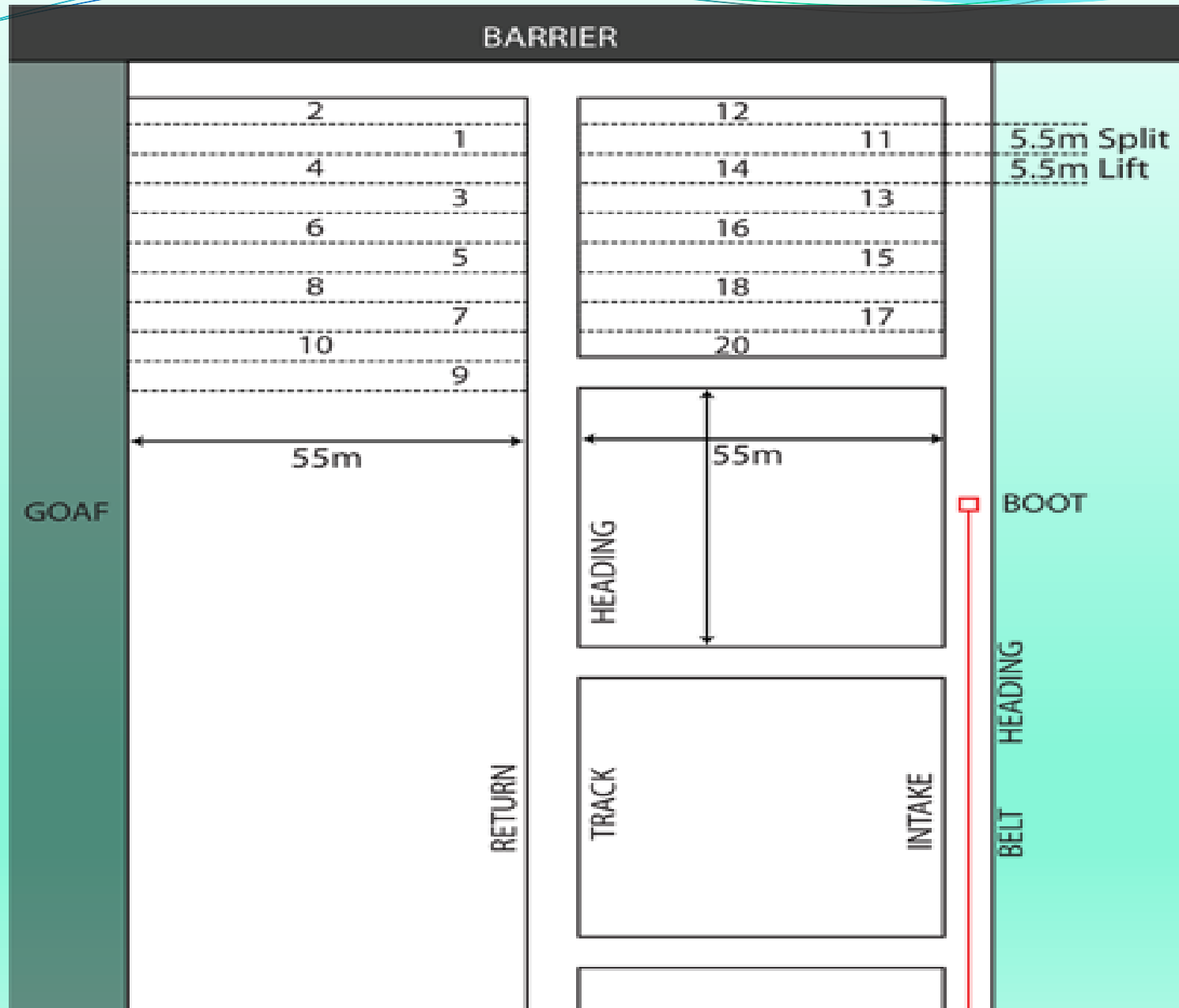
- This method was developed to provide a single working place to extract coal in a stress-relieved area and to utilize the coal seam as support during extraction.
- An overall extraction of 90 per cent was achieved by this system.

WONGAWILLI METHOD

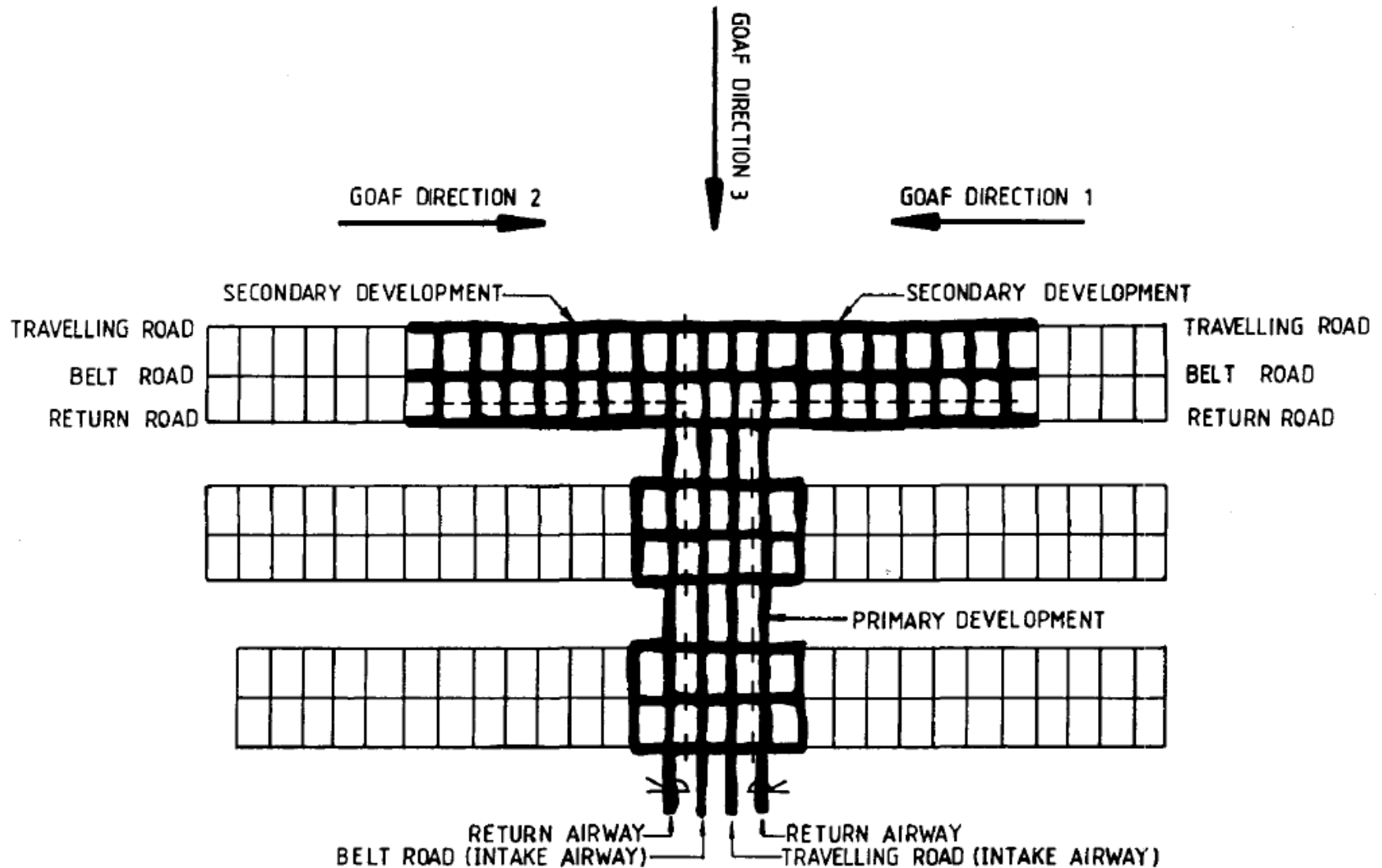
- The main disadvantages of the Wongawilli system area:
- ➤ Excessive floor lift when splitting successive headings in a large panel
- ➤ Difficulties when removing stooks on the return run out of each heading
- ➤ Difficulties with ventilating rib pillar panels when the roof caves completely, thus filling voids in the goaf area and choking the ventilation flow.

WONGAWILLI METHOD

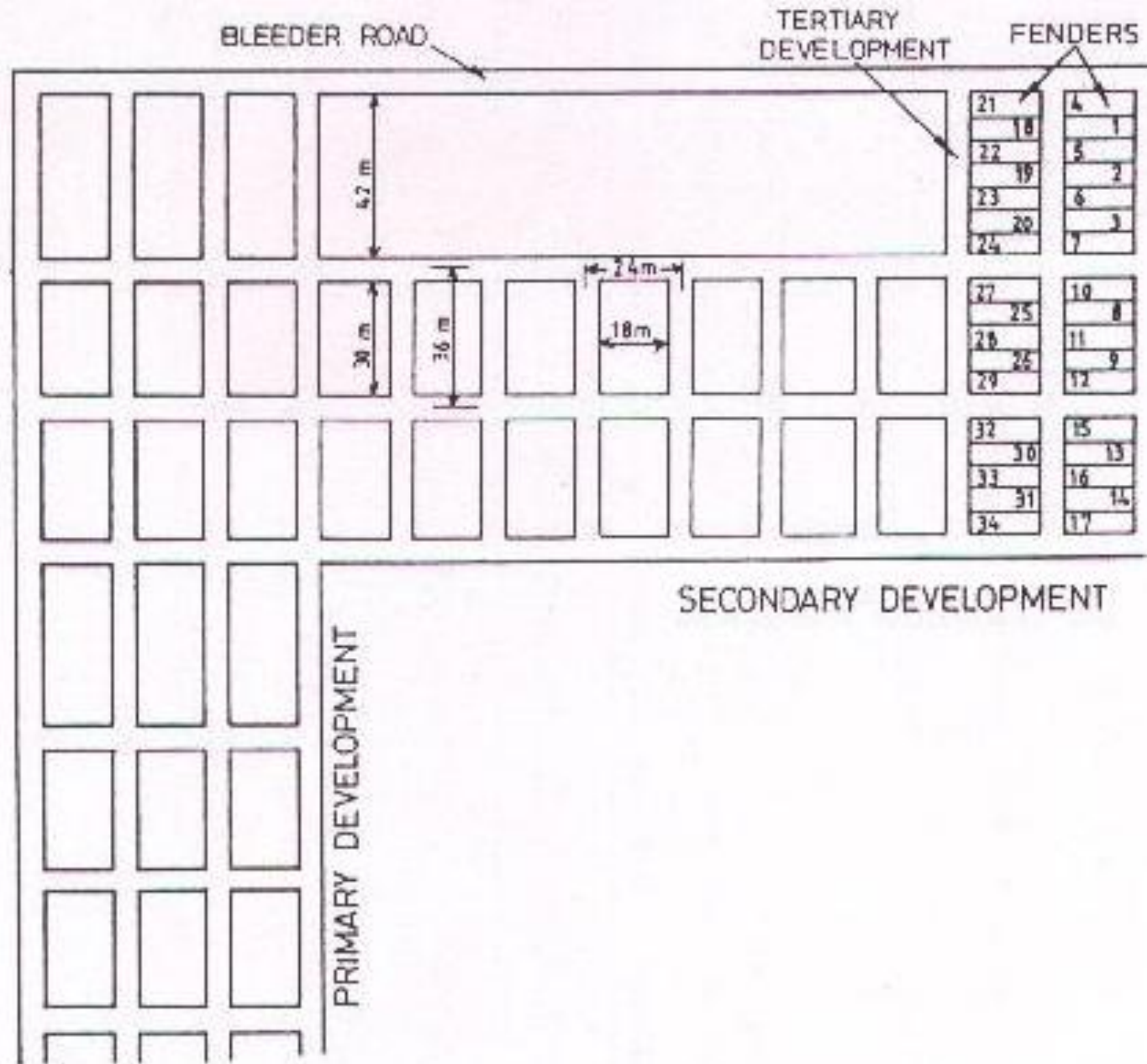
- 1. Provision of a single working place
- 2. Extraction of coal in stress relieved areas
- 3. Maximise extraction of the seam,
- 4. Simple system readily understood by all employees.
- The method provides;
 - a) A straight goaf line
 - b) Simple ventilation system, with air being coursed via the goaf, and
 - c) Working within de-stressed areas.



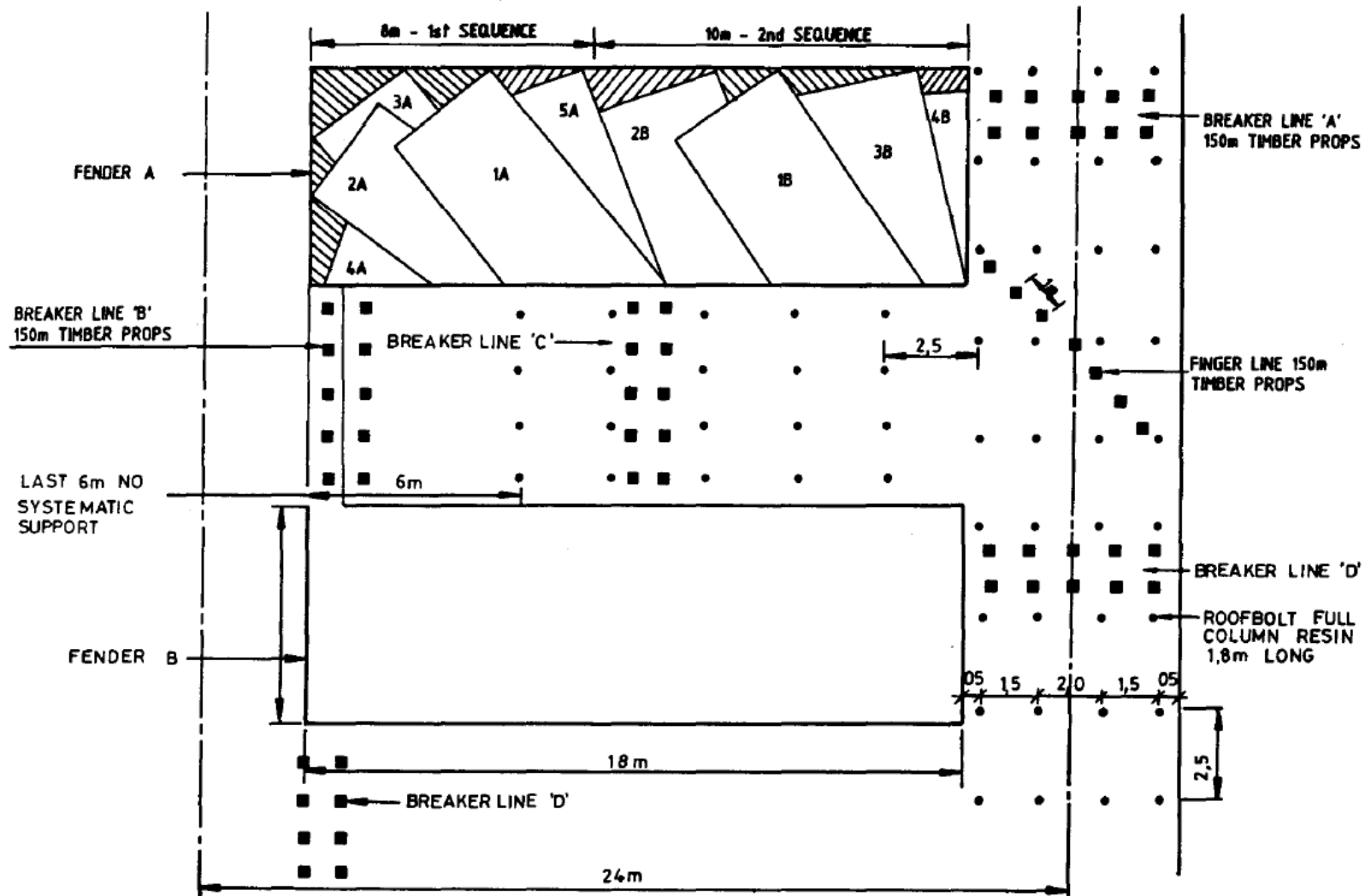
RIB PILLAR EXTRACTION METHOD



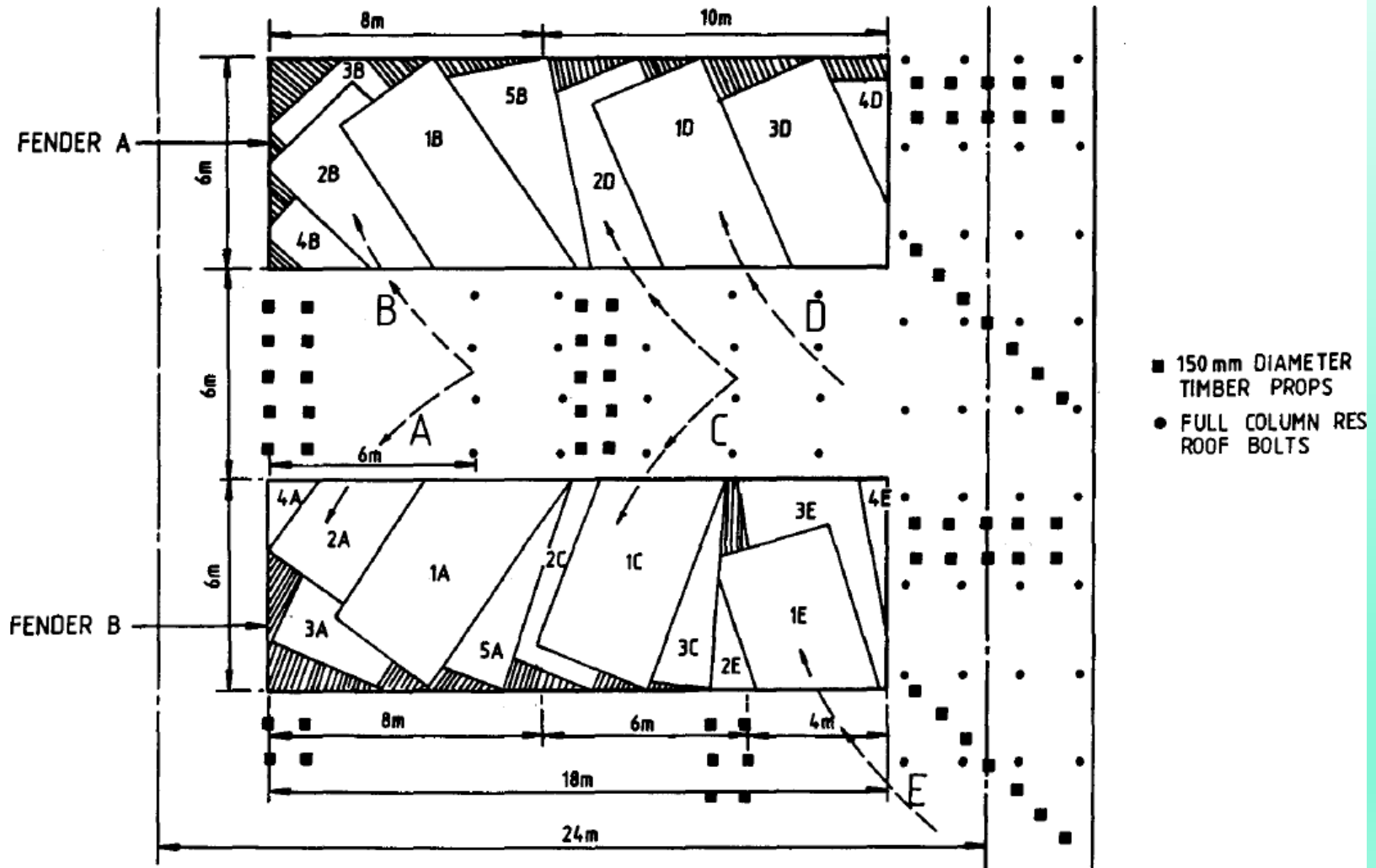
RIB PILLAR EXTRACTION METHOD



RIB PILLAR EXTRACTION METHOD – SYSTEMATIC CUTTING AND SUPPORT SEQUENCE



RIB PILLAR EXTRACTION METHOD – SYSTEMATIC CUTTING AND SUPPORT SEQUENCE



RIB PILLAR EXTRACTION METHOD

- *Panel Layout and Cutting Sequence*
- The area allocated to rib-pillar mining is divided into workable sections, and the primary development normally consists of four roadways from the main development to the limit of the remnant. The two outer roads are utilized as return airways, and the two inner road travelling and conveyor-belt roads. The inner roads also serve as intake airways
- The secondary development consists of three roads, two being intake airways and one a return airway.
- Cross conveyor installations are used to ensure that the tramming distance for shuttle cars is minimized.

PR OJ EC TS	APP 1		APP 2		APP 3		APP 4		APP 5		Total	
	Target	Actual	Target	Actual	Target	Actual	Target	Actual	Target	Actual	Target	Actual
Chirimiri												2,452,820
13.05.02	273,750	463,297	318,825	519,226	404,423	480,129	417,821	523,370	433,403	466,798	1,848,222	31.03.2010
Tandsi												2,034,235
01.09.03	350,000	314,705	437,000	495,625	437,000	543,950	437,000	270,465	437,000	-	2,098,000	07.08.15



- | Cutting Heights | |
|-----------------|--------------|
| • 14CM | : 0.8 - 3.6m |
| • 12CM | : 1.8 - 4.6m |
| • 12HM | : 2.0 - 6.0m |

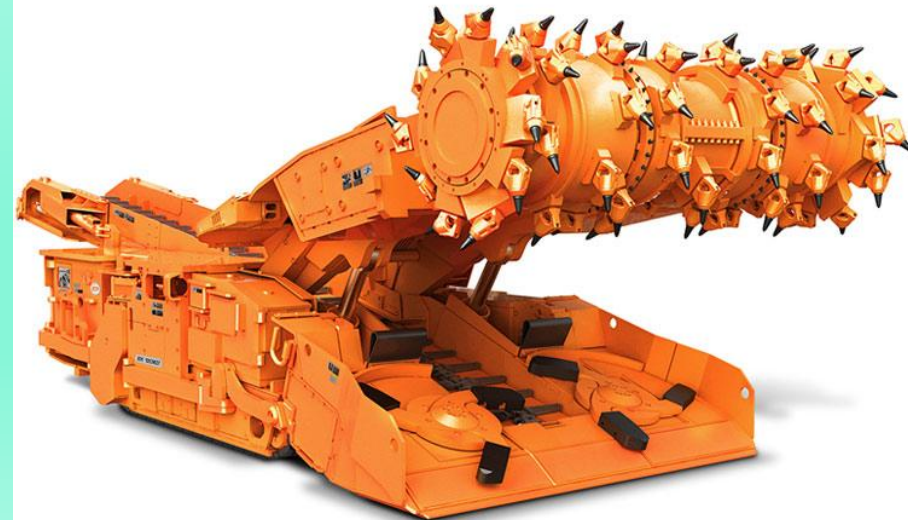
Cutting height is the minimum height required to make way for its own boom. Operating height should be more by at least 200 mm. Min. Cutting height is without scrubber.

CM Selection: Some Broad Guidelines

	14CM15	12CM12	12CM15	12CM27	HM
Operating Range (m)	0.814 – 2.325	1.270 – 3.683	2.16 - 4.6	1.56 / 1.746 – 3.76 / 5	2.0 – 6
Weight (Te)	51.4	59	58	74.5	77
Loading Range (Te/min.)	10 - 28	15 - 29	15 - 27	21 - 38	19 - 35
Conveyor Width (mm)	762	965	762	965	965
Cutting Width (mm)	3500	3300	3300	3500	4115
Drive	AC Opti	AC Opti	DC	AC Opti	AC Opti
Control & Diagnostics	Full JNA	Full JNA	Half JNA	Full JNA	Full JNA
Operating Voltage	1100	1100	1100	3300	3300
Power (Kw)	590	484	545	807	894

Cutting Height (m)

Joy 12 CM 12 D	2.16 to 4.6
Joy 12 CM 12 B	1.37 to 3.68
Joy 12HM31	3.8 to 4.8
Joy 12 CM27E	2.4 to 5.0



Thin Seam Continuous Miners



14CM10-10A Continuous Miner

Thin Seam Continuous Miners



Shuttle Cars

Wide range: from 7 Tonne to 30 tonne



Shuttle Cars

Increasing Coal Hauling Productivity

•21SC 04

- Seam height : low
- Payload : 8 tonnes



•10SC32 AA, A, AB

- Seam height : low to mid
- Payload : 14 tonnes



•10SC32 B, BC

- Seam height : mid
- Payload : Up to 16 tonnes



•10SC32 C

- Seam height : mid to high
- Payload : 18 tonnes



•21SC3AA

- Seam height : low
- Payload : 7.7 tonnes



•10SC32 D

- Seam height : high
- Payload : 30 tonnes

Battery Hauler



Can operate in low seams. Payloads upto 18 tonne. Quiet & cpmfortable, easy battery change outs and low operating cost per tonne.



Diesel / Electric Coal Hauler

Ranging from payloads of 6 tonne and above

Bolters: Versatility is the Key



**Twin Boom
Twin Rig
2.5 – 5.5 m**



**Twin
Boom:
Quad Rig**



**Using the
Temporary
Roof Support**



Side Bolting

Bolters: Versatility is the Key



Thin Seam 1 to 1.9

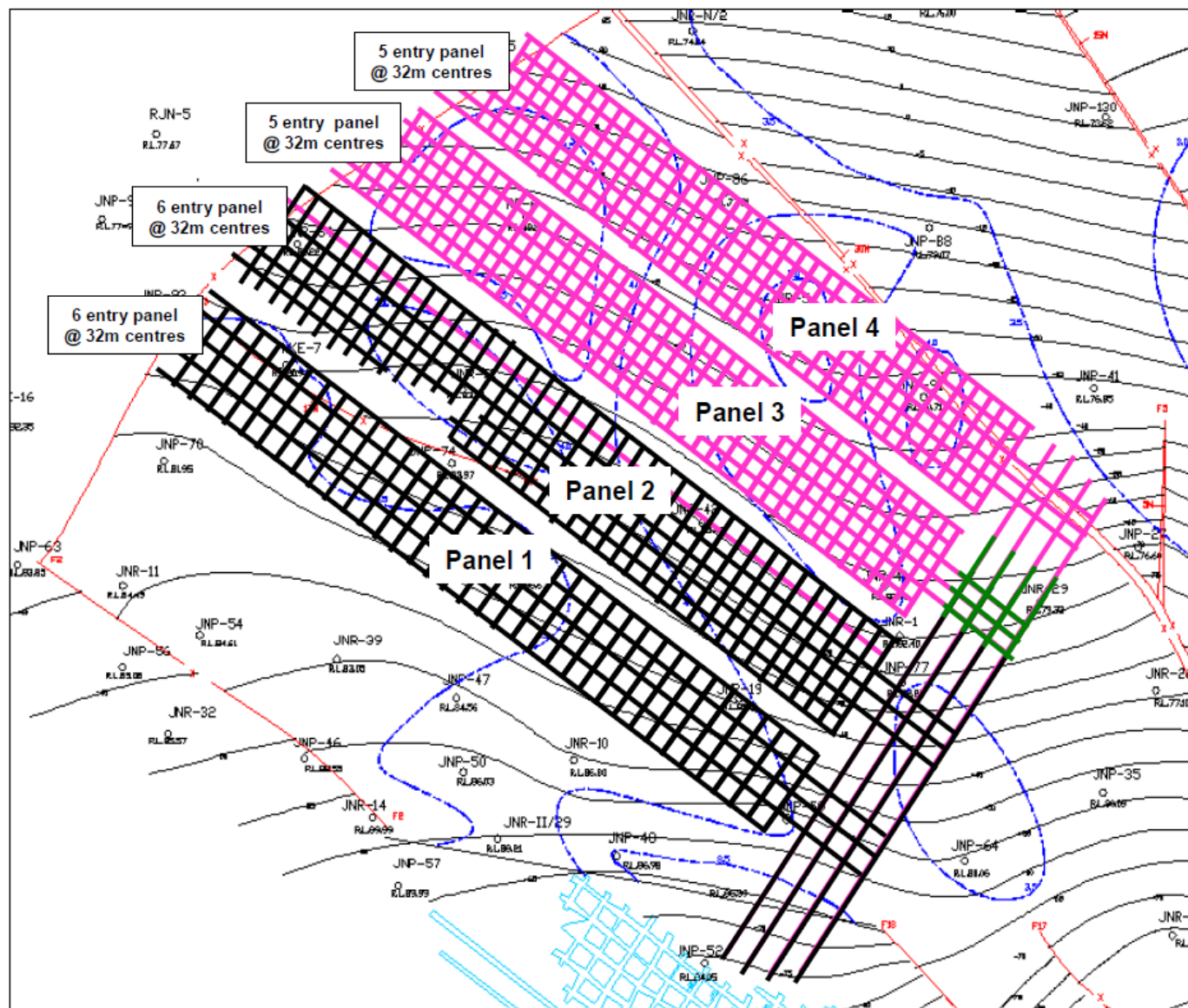


2.3 – 3.3



1.5 – 2.5

Panel Formation & Layout: Jhanjra



S.N	Coalfield	State	Company	Project	Cut-Out Distance	1 st APP Production
1.	Raniganj	West Bengal	ECL	Jhanjra	14 m	631820
2.	Raniganj	West Bengal	ECL	Sarpi	14 m	483906
3.	Chirimiri	Chhattisgarh	SECL	Chirimiri	12 m	463297
4.	Central India	Chhattisgarh	SECL	Sheetaldh ara	7 -14 m	518960
5 #	Godavari Valley	Andhra Pradesh	SCCL	Venkatesh Khani-7	15 m	300240 #